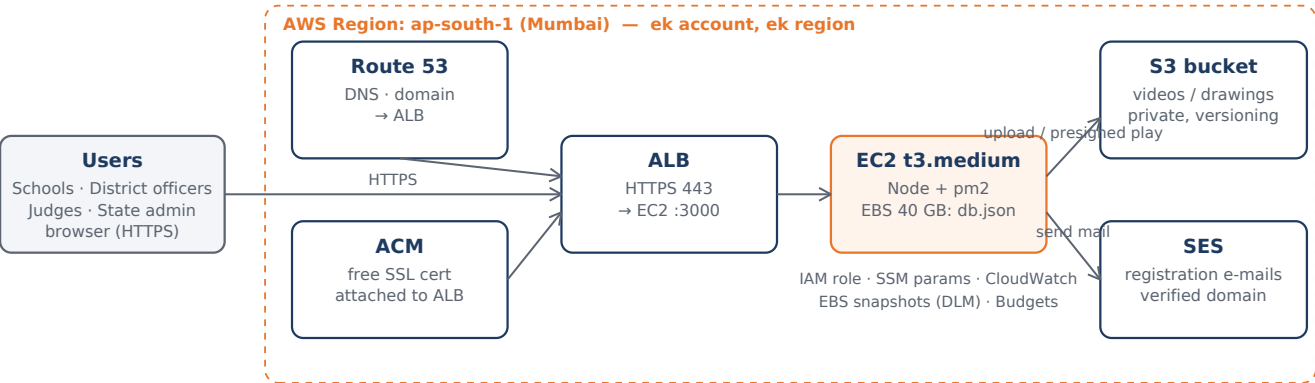


AWS Deployment Plan — sab kuch ek hi AWS account me

Portal (Node/Express, JSON-file DB, video uploads up to 200 MB) ko Railway se hataakar poora AWS Mumbai region me chalana hai, sirf campaign ke ek mahine ke liye (17 Sep – 17 Oct 2026, Grand Showcase 17 Oct). Target: ~1,000+ Delhi govt schools, 13 districts, judges/officials ka Evaluation Desk. Design principle: **ek account, ek region, koi third-party service nahi** — domain, SSL, email, storage, backup, alerts sab AWS ke andar.

Decision	
Region	ap-south-1 (Mumbai) — data India me rehta hai (BLUEPRINT open question #7 ka jawab)
Compute	1 × EC2 t3.medium (2 vCPU / 4 GB) — Node + pm2, app port 3000; koi container/Kubernetes nahi
HTTPS	Application Load Balancer + ACM free certificate (Let's Encrypt/Certbot ki jagah — pure AWS)
Storage	Videos/drawings → S3 private bucket; db.json + code → EBS 40 GB gp3
Email	Amazon SES (Brevo ki jagah) — registration mails; domain Route 53 se verify
Domain/DNS	Route 53 hosted zone; agency domain yahi se manage; NIC subdomain ho to unse CNAME/A record
Secrets	SSM Parameter Store (free) — ADMIN_PASS, SESSION_SECRET etc.; server pe .env nahi
Backup	EBS daily snapshot (Data Lifecycle Manager, 7 din) + S3 versioning
Monitoring	CloudWatch alarms (CPU, disk, ALB 5xx) + AWS Budgets alert Rs. 10,000
Estimated bill	Rs. 9,000 – 11,000 for the month (details in section 5)

1. Architecture



Flow: browser → Route 53 → ALB (HTTPS, ACM cert) → EC2 (Node app). App videos ko S3 me likhta hai aur judges ko playback ke liye short-lived presigned URLs deta hai (files kabhi public nahi hoti — RFP section G consent/PII ke hisaab se). Registration email SES se jaata hai. db.json EBS pe rehta hai, roz snapshot hota hai.

2. AWS services — kya kya banana hai

Service	Kyun / kya karega	Config
Account basics	Root pe MFA, ek IAM admin user, billing alerts on. Sab kaam IAM user se, root se nahi.	IAM Identity Center optional; simple IAM user + MFA kaafi hai
VPC (default)	Default VPC use karo, naya mat banao. 2 public subnets ALB ko chahiye (default me hote hain).	Security groups: ALB-SG (443, 80 from 0.0.0.0/0) · EC2-SG (3000 from ALB-SG only, 22 from apna IP)
EC2	App server. Ubuntu 24.04 LTS, t3.medium. Node 20 LTS, pm2, git. Upload week me load dikhe to t3.large (stop → change type → start, 2 min).	AMI: Ubuntu 24.04 · Instance profile = portal-ec2-role · EBS gp3 40 GB, delete-on-termination OFF
EBS + DLM	Root volume pe data/db.json aur code. Data Lifecycle Manager policy: daily snapshot 02:00 IST, 7 retain.	Tag volume Backup=daily; DLM policy target that tag
S3	portal-media-<unique> bucket, Mumbai. Block all public access ON, versioning ON, SSE-S3 encryption default. Lifecycle: noncurrent versions delete after 14 days.	CORS not needed (uploads go via server). Folder prefixes: reels/, drawings/, exports/
IAM role (EC2)	portal-ec2-role: s3:PutObject/GetObject/DeleteObject/ListBucket sirf is bucket pe; ses:SendEmail; ssm:GetParameter path /portal/*; CloudWatch logs.	Koi access key server pe nahi — role se credentials automatic
ALB + ACM	HTTPS terminate ALB pe. ACM me public cert (DNS validation, Route 53 se ek click). Listener 443 → target group EC2:3000; listener 80 → redirect 443. Idle timeout 300 s (bade uploads ke liye).	Health check path: / (200). Target group deregistration delay 30 s
Route 53	Hosted zone for domain. A/ALIAS record → ALB. Agar domain GoDaddy/others pe hai: nameservers Route 53 pe point kar do (ek baar), phir sab yahi.	Records: portal.<domain> ALIAS ALB · SES DKIM/MAIL FROM records auto
SES	Registration confirmation mails. Identity = domain (Route 53 se DKIM auto). Production access request Day 1 pe hi daal do — sandbox me sirf verified addresses pe mail jaata hai; approval 24 h tak leta hai.	Region Mumbai · MAIL_FROM = noreply@<domain> · SDK: @aws-sdk/client-sesv2
SSM Parameter Store	Env vars: /portal/ADMIN_USER, /portal/ADMIN_PASS (SecureString), /portal/SESSION_SECRET, /portal/S3_BUCKET, /portal/MAIL_FROM. pm2 start script inhe padhta hai.	Free tier (standard params)
CloudWatch	Alarms: EC2 CPU > 80% 10 min, disk > 80% (agent), ALB 5xx > 20/5 min, ALB unhealthy host. SNS topic → apna email/SMS.	CloudWatch agent for disk + app logs (pm2 logs → /var/log/portal)
AWS Budgets	Monthly budget Rs. 10,000 (~\$120): alert at 50%, 80%, 100% forecast.	Free

Jo NAHI lena hai (ek mahine ke liye over-engineering)

RDS/Postgres (5-6k records ke liye JSON file theek hai, snapshot le rahe hain — bas store.js me write-lock lagana hai), ECS/EKS/Docker, Auto Scaling group, CloudFront (judges ki playback S3 presigned URL se direct chalegi; public gallery kholni ho tab jodo), ElastiCache/Redis (single instance, in-memory sessions chalengi), WAF (ALB-SG + rate limit app me kaafi hai), Elastic IP (ALB ke saath zaroorat nahi).

3. Code me kya badlega

Teen files chhoti hain, ek din ka kaam. Local pe test karke phir deploy.

File	Change	Effort
package.json	Add: @aws-sdk/client-s3, @aws-sdk/s3-request-presigner, @aws-sdk/client-sesv2, multer-s3. Remove nothing (Brevo code ko fallback rakh sakte ho, par key nahi dena).	10 min
server.js (multer)	multer.diskStorage → multer-s3 (bucket from env, key = reels/<uuid>.mp4, contentType auto). /uploads/<file> route → 302 redirect to presigned GET URL (expires 1 h). Delete entry → s3.DeleteObject. Range/seek S3 handle karta hai.	3-4 h
src/mail.js	Brevo REST call → SESv2 SendEmail (same subject/body). From = MAIL_FROM. Errors log karo, registration fail mat karo (jaise abhi).	1 h
src/store.js	Write queue/lock: saare writes ek promise-chain me serialize karo + write to db.json.tmp then rename (atomic). Isse 200 schools ek saath save karein to bhi file corrupt nahi hogi.	2-3 h
ecosystem.config.js (new)	pm2 config: name portal, script server.js, instances 1, env from SSM (start script me aws ssm get-parameters-by-path).	30 min
Existing data	Railway /data volume se db.json + uploads/ download → db.json EBS pe, videos aws s3 cp --recursive; db.json me videoFile paths ko S3 keys me map karo (chhota script).	1-2 h

Nginx nahi chahiye — ALB directly Node :3000 pe bhejta hai. Express me `app.set('trust proxy', 1)` aur session cookie secure: `true` lagana (ALB se X-Forwarded-Proto aata hai).

multer-s3 sketch

```
const { S3Client, GetObjectCommand, DeleteObjectCommand } = require('@aws-sdk/client-s3');
const { getSignedUrl } = require('@aws-sdk/s3-request-presigner');
const multerS3 = require('multer-s3');
const s3 = new S3Client({ region: 'ap-south-1' }); // creds from EC2 role
const upload = multer({
  storage: multerS3({ s3, bucket: process.env.S3_BUCKET, contentType: multerS3.AUTO_CONTENT_TYPE,
    key: (req, file, cb) => cb(null, `reels/${Date.now()}-${crypto.randomUUID()}${path.extname(file.originalname)}`),
    limits: { fileSize: 200 * 1024 * 1024 }
  });
});
app.get('/uploads/:key(*)', requireLogin, async (req, res) => { // private playback
  const url = await getSignedUrl(s3, new GetObjectCommand({ Bucket: process.env.S3_BUCKET, Key: req.params.key }));
  res.redirect(url);
});
```

SES sketch (src/mail.js)

```
const { SESv2Client, SendEmailCommand } = require('@aws-sdk/client-sesv2');
const ses = new SESv2Client({ region: 'ap-south-1' });
async function sendMail({ to, subject, html }) {
  try {
    await ses.send(new SendEmailCommand({
      FromEmailAddress: process.env.MAIL_FROM, Destination: { ToAddresses: [to] },
      Content: { Simple: { Subject: { Data: subject }, Body: { Html: { Data: html } } } } }));
  } catch (e) { console.error('mail failed', e.message); } // never block registration
}
```

4. Setup order — 2 din

Day 1: account + infra (aadha din) · Day 2: code + migration + test. Go-live 17 Sep ke liye 16 Sep raat tak DNS switch.

Day 1 · Account

- AWS account → root MFA → IAM user tilak-admin (AdministratorAccess, MFA) → Budgets: Rs. 10,000 alert.
- Region switch: **Asia Pacific (Mumbai) ap-south-1** — har service isi region me banao (SES bhi).
- SES → Verified identities → domain add → 'Publish DNS records to Route 53' → **Request production access** (use case: transactional registration emails, ~1,000 schools). Ye 24 h le sakta hai, isliye sabse pehle.

Day 1 · Network + storage

- Route 53 → hosted zone (domain agency ka hai to registrar pe nameservers Route 53 ke daal do).
- S3 → bucket portal-media-sevasankalp (Mumbai) → Block public access ON · Versioning ON · Lifecycle rule noncurrent 14 d.
- IAM → role portal-ec2-role (trusted entity EC2) → inline policy: S3 (that bucket), ses:SendEmail, ssm:GetParameter*, CloudWatchAgentServerPolicy.
- SSM Parameter Store → /portal/ADMIN_USER, /portal/ADMIN_PASS (SecureString, strong), /portal/SESSION_SECRET (64 random chars), /portal/S3_BUCKET, /portal/MAIL_FROM.
- Security groups: portal-alb-sg (80, 443 from anywhere) · portal-ec2-sg (3000 from portal-alb-sg; 22 from my IP only).

Day 1 · Server

- EC2 → Launch: Ubuntu 24.04, t3.medium, key pair, portal-ec2-sg, IAM role portal-ec2-role, gp3 40 GB, tag Backup=daily, Name=portal.
- SSH → install Node 20 (nodesource), pm2, git, awscli, CloudWatch agent → git clone repo → npm ci → pm2 start ecosystem.config.js → pm2 save && pm2 startup.
- EC2 → Lifecycle Manager → policy: EBS snapshot daily 20:30 UTC (02:00 IST), retain 7, target tag Backup=daily.
- ACM → Request public cert portal.<domain> → DNS validation → Create records in Route 53 → Issued (5–30 min).
- EC2 → Load Balancers → ALB portal-alb (internet-facing, 2 subnets, portal-alb-sg) → target group portal-tg (HTTP 3000, health /) → listener 443 (ACM cert) → forward; listener 80 → redirect 443. Attributes: idle timeout 300 s.
- Route 53 → A record portal.<domain> ALIAS → ALB. Test https://portal.<domain>.
- CloudWatch → alarms (CPU, disk, ALB 5xx, unhealthy hosts) → SNS topic → email confirm.

Day 2 · Code + data

- Local: multer-s3, SES, store.js lock, trust proxy → test against real bucket (aws sso/keys local pe sirf test ke liye).
- Railway se db.json + uploads/ download → aws s3 cp uploads/ s3://bucket/reels/ --recursive → mapping script se db.json update → scp to EC2 data/.
- git push → EC2 pe git pull && npm ci && pm2 reload portal.
- Smoke test: register school (SES mail aaya?), 150 MB upload, playback seek, judge scoring, CSV export, /staff login, admin password changed.
- Railway app ko delete mat karo 2-3 din — rollback ke liye rakho; DNS switch hone ke baad hi band karo.

EC2 bootstrap (user-data ya SSH ke baad)

```
sudo apt update && sudo apt -y install git unzip
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash - && sudo apt -y install nodejs
sudo npm i -g pm2
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o awscliv2.zip && unzip -q awscliv2.zip && sudo .
git clone https://github.com/ganbjhs/StudentPortal.git ~/portal && cd ~/portal && npm ci --omit=dev
# env from SSM at start (ecosystem.config.js runs this)
export $(aws ssm get-parameters-by-path --path /portal --with-decryption --region ap-south-1 \
  --query 'Parameters[][Name,Value]' --output text | sed 's#/portal/##' | awk '{print $1="$2}" | xargs)
pm2 start server.js --name portal && pm2 save && pm2 startup systemd -u ubuntu --hp /home/ubuntu
```

5. Estimated cost — ek mahina, Mumbai region

Assumptions: 1,000 schools × ~4-5 entries × ~80 MB = ~400-500 GB media; judges/officials ~400 GB playback (data transfer out); Rs. 84/USD. On-demand pricing (Sep 2026 list rates, approx). Naya account ho to free-tier credits mil sakte hain — check karo.

Item	Sizing	USD / month	Rs. / month
EC2 t3.medium	730 h on-demand	~33	~2,800
EBS gp3 40 GB	root + db.json	~4	~350
EBS snapshots	~7 × incremental, ~40 GB base	~2	~200
ALB	730 h + low LCU	~20	~1,700
S3 Standard	500 GB + PUT/GET requests	~14	~1,200
Data transfer out	400 GB (first 100 GB free) @ ~\$0.109/GB	~33	~2,800
Route 53	1 hosted zone + queries	~1	~100
SES	~2,000 mails	<1	~20
CloudWatch / SSM / Budgets / IAM / ACM	basic	~2	~150
Total		~\$110	~Rs. 9,300 (range 9,000-11,000)

Sabse bada variable data transfer out hai — agar Grand Showcase pe public gallery kholte ho aur hazaron log video dekhte hain to ye badhega; tab CloudFront jodna sasta padta hai (Mumbai CloudFront ~\$0.109/GB same, par 1 TB free/month). Domain registration alag (~\$13/yr agar naya lena hai).

6. 17 Oct ke baad — bill band karne ki checklist

Ek mahine ka project hai, isliye shutdown plan bhi zaroori hai warna bill chalta rahega. Order matter karta hai.

- Final exports le lo: admin CSV/participation report, saare selected reels (aws s3 sync s3://bucket/reels/ ./archive/), db.json copy — RFP section F 'digital archive' yahi hai. Agency/DoE ko hard-drive ya S3 Glacier me do.
- Archive rakhna ho: S3 lifecycle → Glacier Deep Archive (~Rs. 0.10/GB/month, 500 GB = ~Rs. 50/month). EC2/ALB band, sirf bucket rahe.
- Delete order: ALB → target group → EC2 terminate (EBS snapshot final le lo pehle) → DLM policy → EBS volume → ACM cert → CloudWatch alarms → SNS. S3 bucket sabse aakhri me (ya Glacier me rakho).
- Route 53 hosted zone rakho agar domain aage bhi use hoga (Rs. 50/month), warna delete. SES identity rakh sakte ho, free hai.
- Billing console → Cost Explorer next month check karo ki kuch chhoota nahi (EBS orphan volumes aur snapshots aksar chhoot jaate hain).

7. Dhyan rakhne wali baatein

- **SES sandbox:** production access approve hone tak sirf verified emails pe mail jaayega. Day 1 pe request daalo; approve na ho to registration phir bhi chalega (mail skip hota hai, jaise abhi).
- **NIC subdomain (edudel.nic.in):** agar DoE chahta hai to NIC ko ALB ka DNS name dena hoga CNAME ke liye — ye process hafta le sakta hai; parallel me agency domain pe live rakho.
- **Single instance:** ALB ke peeche ek hi EC2 hai — reboot pe 1-2 min down. Ek mahine ke liye acceptable; upload week (pehla hafta) me deploy/reboot raat me karo.
- **200 MB uploads:** ALB idle timeout 300 s set karna zaroori hai; slow school connections pe 200 MB ~5-10 min le sakta hai — app me clear progress bar/timeout message rakho.
- **JSON DB:** store.js write-lock ke bina concurrent saves me corruption ka risk hai — ye code change skip mat karo. Snapshot se 24 h purana data hi wapas aayega, isliye roz raat manual `aws s3 cp data/db.json s3://bucket/exports/db-$(date +%F).json` cron bhi laga do (free, 5 min).
- **Admin password:** seeded admin/admin123 ko pehle login pe hi badlo; SSM me strong value rakho.
- **PII:** student names, class, consent flags hain — bucket public kabhi mat karo, presigned URL 1 h ka rakho, CSV exports S3 exports/ prefix me hi rakho, server pe nahi.